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Shock tube investigations of ignition delays of *n*-butanol at elevated pressures between 770 and 1250 K

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Abstract

The ignition delays of *n*-butanol, a potential bio-fuel candidate, have been determined in a high-pressure shock tube. Conditions behind the reflected shock are approximately between 10–42 bar and 770–1250 K. To our knowledge, the ignition delay measurements of butanol at these high pressures are the first of their kind. CH emission and pressure time histories have been probed to determine ignition delay times for all experiments. For stoichiometric fuel–air-mixtures the influence of the temperature and pressure has been characterized. Interestingly the experimental data deviate from the Arrhenius behavior for temperatures lower than 1000 K. This is in contrast to simulation results which have been obtained by employing the simulation tool CANTERA with different reaction mechanisms applying the typical assumption of isochoric conditions. It has been found out that a positive pressure and temperature gradient behind the reflected shock has a significant influence on the ignition delay below 1000 K causing a pronounced decrease in the ignition delay times. This change of the conditions behind the reflected shock is attributed to the shock attenuation and probably from pre-ignition. Including the measured pressure gradients and assuming an isentropic compression behind the reflected shock, the simulation data and the experimental results show a same trend in the temperature dependence of the ignition delay. Nevertheless, striking differences between experiment and simulation persist especially for higher pressures. By performing sensitivity analysis at different temperatures some critical reactions could be identified and their role under our experimental conditions is discussed. In summary it can be stated that the employed reaction mechanisms may not be fully applicable to high-pressure conditions and it seems plausible that the lack of more detailed low temperature fuel specific reactions could be the probable cause for the discrepancies which calls for detailed investigations at elevated pressures.
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1. Introduction

The investigation of combustion characteristics of hydrocarbon fuels is still a necessary task today. Due to the foreseen scarcity of fossil oil, the development and investigation of new biofuels for their ignition properties is the main focus of the current work. Besides ethanol, butanol is

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considered a potential bio-fuel candidate and possible production routes have been investigated in former studies [1–3]. In comparison, butanol is less hygroscopic and has a higher energy density than ethanol. These advantage is one of the motivation for investigating the ignition of butanol under engine relevant conditions.

Ignition delays of butanol oxidation have been previously reported in the literature. In Ref. [4] the oxidation of four isomers of butanol is investigated. A kinetic mechanism has been developed for high temperatures for predicting ignition delays and this has been validated with shock tube experiments. The reported pressures and temperatures behind the reflected shock range from 1 to 4 bar and 1200 to 1800 K, respectively. Another recent shock tube study concerning the oxidation of n-butanol has been presented in Ref. [5] where results for pressures between 1 and 8 bar and temperatures between 1100 and 1800 K have been reported. The developed kinetic model has been used to simulate the shock tube experiments and for lower pressures a good agreement is obtained. Interestingly, for the highest pressure investigated (8 bar) in Ref. [5], the ignition delays predicted by the simulation are longer than that observed in experiment especially for high equivalence ratios. In the same study, simulations are also compared with results from a jet stirred reactor. These experimental results have been reported for 1 and 10 bar for different equivalence ratios [6]. The comparison of kinetic simulation [5] and experiments from Ref. [6], for temperatures between 800 and 1100 K shows an overall good agreement. In addition to pure butanol the oxidation kinetics of a butanol-surrogate gasoline 85:15 blend is also investigated in a jet stirred reactor at 10 bar and from 770 to 1120 K in a separate study [7].

Typically, experimental investigations are needed to validate kinetic mechanisms and to determine Arrhenius type correlations for the ignition delay time. However, the range of conditions that can be found in literature needs to be extended for describing the ignition process in combustion engines where significant higher pressures and low temperatures are desired. Accurate and reliable experimental data over extended temperature and pressure conditions is therefore needed which is a further motivation for this work. One common example is the behavior of n-heptane which shows a distinct negative temperature coefficient (NTC-behavior) in the intermediate temperature region. This especially concerns the low temperatures oxidation chemistry which has a very complicated network of temperature and pressure dependent reaction pathways. In this case the ignition can not be described by a simple Arrhenius approach and *ad hoc* reactions mechanisms developed for high temperatures and low pressures fail to describe extended temperature and pressure conditions. Even if no NTC-behavior is observed, significant errors might occur when low pressure data are extrapolated to higher pressures since pressure effects are still not well understood for novel fuels. Therefore the aim of this work is to characterize the oxidation of *n*-butanol with respect to its ignition delay under extended pressure and temperature ranges for attaining engine like conditions. This will further help to develop and validate comprehensive mechanisms for combustion under extended conditions.

2. Experimental facility

For performing experiments a high-pressure stainless steel shock tube has been used. This shock tube has an inner diameter of 140 mm and a total length of about 15.5 m. The driven section with a length of 11 m is separated from the driver section by a double diaphragm chamber housing two aluminum diaphragms. For investigations of fuels with low vapor pressures the whole facility can be resistively heated up to 200 °C. At this temperature the maximum working pressure of the shock tube is 950 bar. Gas mixtures have been prepared in a separate mixing vessel of 260 l volume. Care has been taken that homogenous mixing of fuel and air (Praxair, 79.5% N_2, 20.5% O_2) is obtained which is monitored by measuring the pressure after complete vaporization of the fuel injected in the previously evacuated mixing vessel and subsequent filling of the vessel with air. The error in the equivalence ratio due to pressure measurement uncertainties has been estimated to be less than 1.5%.

Four piezoelectric gauges (Kistler, model 603B and 6005) have been used for pressure measurements and for the determination of the shock speed. The ignition delay time has been determined by pressure gauges located at several positions (10, 510, 1010 and 2510 mm) from the end wall. In addition, the CH emission at 431.5 nm has been selected by a narrow-band pass filter and measured with a Photomultiplier (EMI), whereby the optical access is also located 10 mm from the end wall. All data has been recorded with a sampling rate of 2.5 MHz.

Before each experiment the shock tube is evacuated below 2×10^{-5} bar. Initial pressures in the driven and driver section range from 0.2 to 2.7 bar and from 30 to 90 bar, respectively. Mixtures of helium and argon or helium and air have been used as driver gases. In order to guarantee the complete vaporization of butanol, the entire facility has been heated up to 80 °C and the mixing vessel up to 100 °C. Fuel–air-mixtures have been allowed to mix for at least 1 h in the shock tube. Longer homogenization times have been tested and no differences in the ignition results have been observed.

It is important to note that only the pressure and the temperature behind the reflected shock are the relevant criteria for determining ignition delays.

Since the pressure can directly be measured, the temperature behind the reflected shock needs to be calculated. This is done using the simulation program KASIMIR [8,9], which is calculates the complete inviscid, one-dimensional shock tube process including equilibrium real gas effects like vibrational excitation and chemical reactions. Input parameters here are the shock speed and the initial conditions in the driven section of the shock tube. The shock speed deduced from the pressure signals is extrapolated for the end wall position assuming constant shock attenuation between the measuring positions. The resulting calculated pressure is compared to all measured values and the differences obtained are within the measurement uncertainty of the pressure probes. Nevertheless uncertainties in the shock velocity and initial temperatures lead to an uncertainty of ±10 K for the temperature behind the reflected shock.

For validating the experimental setup and the data analysis procedure, reference experiments with stoichiometric n-heptane–air and stoichiometric methane–air experiments for pressures ranging from 13 to 38 bar and temperatures ranging from 700 to 1300 K have been performed in an earlier study [10]. For both fuels an excellent agreement to literature data has been obtained for the entire temperature and pressure range investigated.

Ideally the conditions behind the reflected shock are constant with time which is not the case for a real shock tube process [11]. Due to shock attenuation caused by the boundary layer behind the incident shock and non ideal bursting of the diaphragms the pressure and temperature behind the reflected shock, rise with time. This effect is in contrast to the typical assumption of isochoric conditions in a shock tube for kinetic simulations. For shock tubes with large inner diameters, the effect of the boundary layer is minimized and shock attenuation is reduced. In fact the observed facility related shock attenuation is less than 1% in all experiments due to the relatively large inner diameter of 140 mm for the current shock tube. However, the resulting pressure and temperature gradients have an influence on the ignition process at lower temperatures. The pressure transducers were shielded with a layer of high temperature silicone and the pressure gradients were recorded for each transducer in all experiments. The shielding protects the pressure transducer from heat loads of the shock heated gas which would change the pressure signal and lead to significant errors for the pressure gradient. With respect to the measured pressure gradients the temperature gradients can be estimated by assuming an isentropic compression behind the reflected shock [11], whereby γ describes the isentropic exponent (Eq. (1)).

$$\frac{d}{dt}\left(\frac{T}{T_0}\right) = \frac{\gamma - 1}{\gamma} \cdot \frac{d}{dt}\left(\frac{p}{p_0}\right) \tag{1}$$

The uncertainty of the measured pressure gradient $d(p/p_0)/dt$ is approximately ±1% per ms resulting in an uncertainty of ±0.25% per ms for the temperature gradient $d(T/T_0)/dt$. A typical plot of pressure and temperature gradients versus the correlation parameter $Ma_s^{0.5} p_1^{-0.14}$ (obtained from Ref. [11]) is depicted in Fig. 1. Here Ma_s denotes the incident shock Mach number and p_1 the initial pressure of the driven section. For comparison, experiments have been performed not only with fuel–air-mixtures but under similar conditions with air as driven gas. The resulting pressure gradients are in the same order of magnitude indicating that the measured pressure gradient may not be influenced by chemical reactions but due to shock attenuation as discussed earlier.

The ignition delay time is defined as the time difference between the arrival of the incident shock at the end wall and a strong increase of pressure due to the combustion. However, one may find local ignition areas due to small inhomogeneities or perturbances leading to local pre-ignition [12–14]. These local combustion processes lead to a slight increase of the pressure and temperature in the unburnt mixture. At high temperatures, the time difference between these pre-ignitions and the main ignition of the fuel–air-mixture is very short and can hardly been observed although this difference can become significant at low temperatures. Figure 2 depicts the ignition process of a stoichiometric butanol–air-mixture at 880 K and 19 bar where t_{pre} describes the beginning of the local pre-ignitions defined as the time difference between the shock reflection at the end wall and the time where the pressure increase is stronger than from shock attenuation. The onset of the main ignition is described by t_{ign}

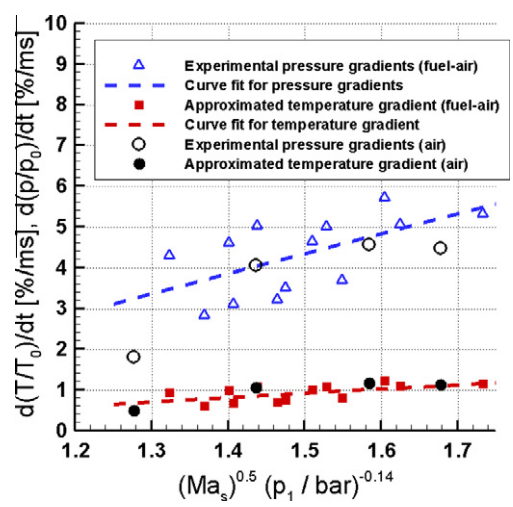

Fig. 1. Experimental pressure and temperature gradients.

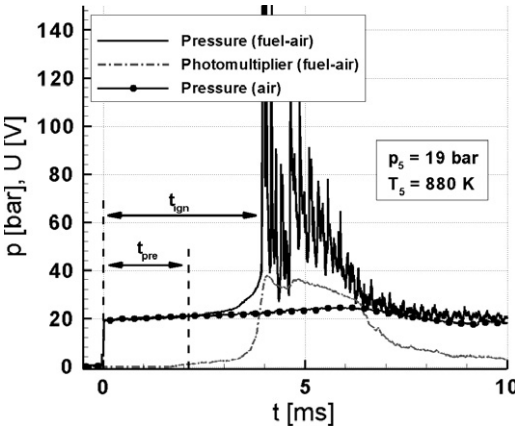

Fig. 2. Definition of the ignition delay time.

indicated by a strong increase in pressure. The ignition process is also detected by the light emission profiles detected by a photomultiplier, and the main ignition is defined as the time of the strongest increase of the signal. The difference between the ignition delay time determined from the pressure probes and the photomultiplier is typically less than 3%.

For low temperature investigations the ignition delay time can amount several milliseconds depending on the type of fuel. Therefore reference experiments with air as driven gas have been performed for estimating the available measuring time. For decreasing shock speeds the available measuring time increases and for optimized conditions measuring times longer than 8 ms have been achieved in this study [10]. The available measuring time is therefore defined as the arrival of the incident shock at the end wall and a significant change in pressure at the end wall due to shock or expansion waves arriving at the end wall. A typical pressure history of an experiment with air as driven gas is shown in Fig. 2. It is important to note that the available measuring time is longer than the ignition delay time and as stated before, the pressure gradient before pre-ignition is only caused by facility effects and not the result of chemical reactions.

3. Simulations

The simulations of the ignition delay times have been carried out using the open source code CANTERA [15,16]. Following the initial physical and boundary conditions of the shock tube the chemical reactions within an isochoric homogeneous reactor have been calculated. The governing equations for such a homogeneous reactor are the conservation equations for the mass fractions Y_i of the species i and the temperature T which are given as

$$\rho \frac{dY_i}{dt} = \dot{m}_i, \tag{2}$$

$$\rho c_v \frac{dT}{dt} = -\sum_{i=1}^{n} u_i \dot{m}_i \tag{3}$$

where $\dot{m}_i$ is the chemical source term which is the rate of change of mass and u_i is the energy of species i.

Furthermore, we investigated the effect of shock attenuation by setting the pressure change behind the reflected shock as boundary condition. Temperature gradients due to these effects are then determined assuming an isentropic compression (Eq. (1)).

As in the experiment, the simulated ignition of the fuel–air-mixture can be determined by a strong increase in pressure and temperature. For the present case, we chose to define the ignition time as the time of maximum pressure and temperature rise. The two different mechanisms selected for the current simulations are denoted as *mech1* [4] and *mech2* [5] discussed later in the text.

4. Results and discussion

Ignition delay times have been measured in the range of 10–42 bar and 770–1250 K. For all experiments stoichiometric ($\Phi = 1$) butanol–air-mixtures have been used. As expected the ignition delay time decreases for higher pressures and higher temperatures (Fig. 3). In the high temperature regime (above 1000 K) the ignition delay time plotted versus temperature can be expressed using an Arrhenius correlation (Eq. (4)), where E_A is the activation energy and A the pre-exponential fac-

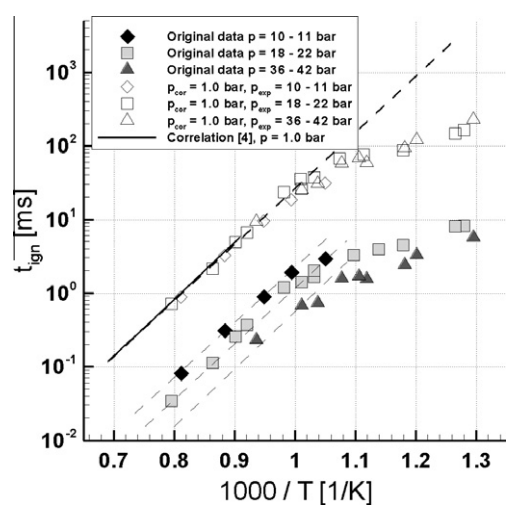

Fig. 3. Ignition delay times for stoichiometric *n*-butanol–air-mixtures.

tor. Based on this approach (Eq. (4)) the pressure dependence of the ignition delay can correlated employing Eq. (5) where the experimental data can be best fitted using $x = -1.0$ with an uncertainty of ± 0.1 which also corresponds to the scatter in the experimental data

$$t_{\text{ign}} = A \cdot p^x \cdot e^{E_a/RT} \tag{4}$$

$$t_{\text{cor}} = t_{\text{exp}} \left(\frac{p_{\text{cor}}}{p_{\text{exp}}} \right)^x \tag{5}$$

Using the pressure dependence relation, present high-pressure results can be compared to the lower pressure data reported in the literature. Figure 3 compares our results with those from Ref. [4] which show good agreement above 1000 K for a reference pressure of 1 bar. This indicates that ignition delay results at different pressures can be compared using a correlation as given in Eq. (5). It is important to note that the correlation proposed in Ref. [4] gives a factor of $x = -0.85 \pm 0.1$ for the pressure dependence which results in longer ignition delay times at higher pressures according to Eq. (4).

Results from the two different mechanisms are compared with the present experimental results correlated with Eq. (5) in Figs. 4 and 5. At 10 bar (Fig. 4) the agreement between all data for temperatures above 1000 K is good and resulting ignition delay times for both mechanisms agree well. However it is important to note that *mech2* gives a slight better agreement with current experiments. Below 1000 K the experimental results deviate from the Arrhenius behavior which is not described by the isochoric simulation. To some extent, this effect which has already been observed for different fuels, is described to be an

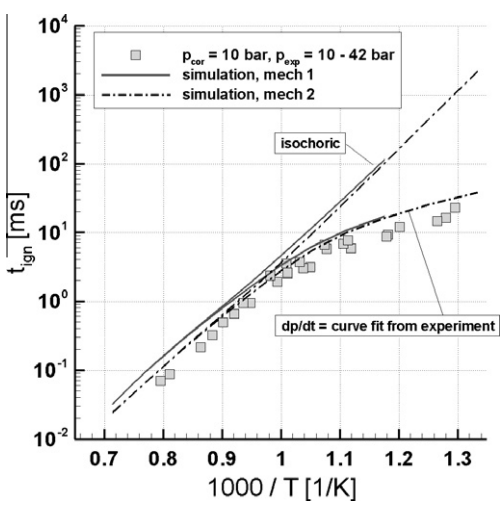

Fig. 4. Simulated and experimental ignition delay time for lower pressure (10 bar).

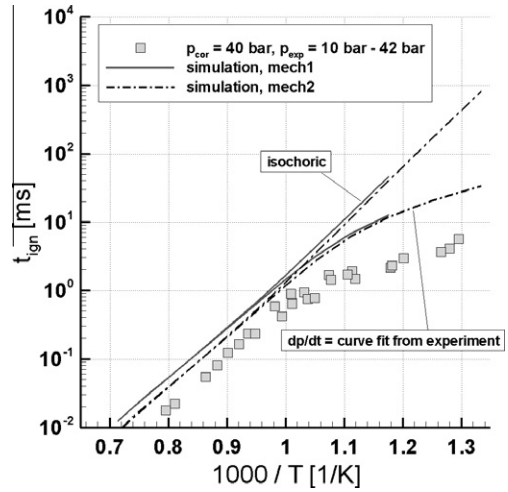

Fig. 5. Simulated and experimental ignition delay time for higher pressure (40 bar).

effect of the pressure and temperature gradients behind the reflected shock [17]. Consequently, the pressure and temperature gradients have been included in the simulation assuming an isentropic compression. A curve fit through the experimental determined pressure gradients has been used for the different conditions and gradients (refer Fig. 1). In this case the simulation results show the same behavior as the experiments, e.g. the ignition delay time data deviate from the Arrhenius behavior for temperatures below 1000 K. Nevertheless, the ignition delay times predicted from the simulation are longer than in experiment. This effect becomes more distinct for higher pressures (Fig. 5). A significant difference between simulation and experiment is observed in the low temperature region, but also for temperatures above 1000 K the ignition delay time resulting from the simulation is longer than that observed in the experiment.

For temperatures above 1000 K the ignition delay is not significantly affected by pressure and temperature gradients so that the discrepancies between simulation and experiment at high pressures likely result from the reaction mechanism employed. In order to discuss the role of the relevant reactions responsible in describing the ignition process of butanol under our experimental conditions, a sensitivity analysis has been performed by adopting the reaction mechanism *mech2* from Ref. [5]. The main purpose of this sensitivity analysis is to identify critical reactions for further studies. When rate constants for certain sensitive reactions are not well known, it would be important to study these reactions separately in order to provide accurate rate coefficients that would facilitate the refinement of a reaction model. Although both reaction mechanisms

(*mech1* and *mech2*) give similar results for ignition delays above 1000 K and for pressures close to 1 bar, either of the mechanisms cannot sufficiently predict our current experimental results at pressures of 40 bar. Furthermore, it is important to note that in Ref. [4] it has already been discussed that the kinetic mechanism (*mech1*) may not be suitable for temperatures below 1000 K and that it fails to capture pressure dependence above a few bar especially because of the simplification of certain estimated rate coefficients and probably neglecting the fall-off effects for uni-molecular dissociations. For this reason the sensitivity analysis has been performed adopting mechanism *mech2* which might be comparable for our conditions although might still lack detailed low temperature reactions. Since the mechanism already includes some reactions of alkylperoxy and hydroperoxyalkyl species, it is beyond the scope of this paper to identify how detailed the low temperature mechanism is. As seen in Fig. 5, *mech2* predicts shorter ignition delays which are closer to the experimental results as compared to *mech1*. However, experimentally obtained ignition delays are much shorter from either of the predictions at pressures close to 40 bar even in the high temperature regime although the deviations start increasing with decreasing temperatures.

This difference between experimental results and simulation could be a result of high uncertainties in certain estimated rate coefficients, e.g. the HO_2 + butanol rate coefficient in *mech1* is estimated from the analogy of alkanes, and also inability of the mechanisms to capture pressure effects since most of the experimental data employed for model validation lies at much lower pressure (~8 bar). To our knowledge, the current experimental results for butanol ignition are at the highest pressures ever investigated.

Figure 6 depicts the sensitivity analysis at three different temperatures for 20 bar to discuss the role of elementary reactions for the ignition delays. As expected, the small radical and $H + O_2$ chemistry dominates at high temperatures (above 1450 K). This is due to the thermal decomposition of *n*-butanol through the C–C bond scissions forming small radicals and due to the reactions of small radicals with species such as HO_2 which is also one of the abundant species under these conditions. It is also no surprise that the fuel specific reaction of $HO_2 + n\text{-}C_4H_9OH \Rightarrow H_2O_2 + C_4H_8OH\text{-}1$ remains most dominant around 1050 K only to compete with the thermal decomposition of *n*-butanol at temperatures around 1400 K. The thermal decomposition of H_2O_2 from the above reaction eventually sets in providing the important chain branching species OH. This is also shown in the declining role of this reaction as temperatures are raised from 1050 to 1400 K.

At very high temperatures, the important OH forming reaction is $H + O_2 \Rightarrow O + OH$ which

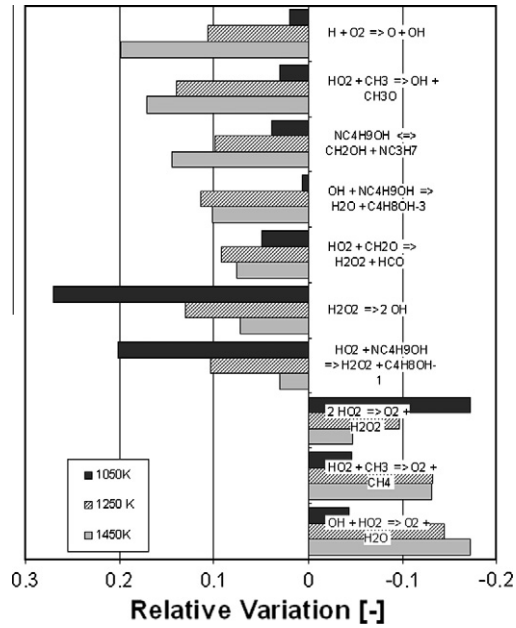

Fig. 6. Sensitivity analysis of ignition delay, mechanism *mech2*, 20 bar, 3% *n*-butanol, 19.9% O_2. Relative variations have been obtained by multiplying the rate constant of each elementary reaction step by a factor of 2.

clearly supports the trend of its dominance as temperatures increase. In addition to this, the sensitivity analysis also reveals the important role of other reactions of HO_2 with species such as OH and CH_3 ($OH + HO_2 \Rightarrow O_2 + H_2O$ and $HO_2 + CH_3 \Rightarrow O_2 + CH_4$) showing a negative influence on the ignition delays. The other relevant reaction and an important source of OH is the reaction of $HO_2 + CH_3 \Rightarrow OH + CH_3O$ which shows an increased influence on ignition delays at elevated temperatures as also discussed in Ref. [5]. This becomes clear because the methyl radicals available for this reaction are mostly produced at high temperatures.

Although, the type of reactions and their role at different temperature regimes becomes distinct from the above sensitivity analysis, the reasons for the large discrepancies between the experiments performed at high pressures and the available comprehensive and detailed mechanisms available in the literature are difficult to trace. In certain cases, such as those discussed in Ref. [4], a probable cause could be the oversimplification by using estimated abstraction rate coefficients for the fuel + HO_2 reaction based on alkanes. This reaction itself has not been well characterized and the deviation of rate coefficients for the same reaction in Refs. [4,5] is about a factor of 20 for our experimental conditions which clearly calls for careful and detailed studies of this reaction over a wide temperature and pressure range. Since butanol has been

not well studied for its elementary kinetics especially for its initiation oxidation chemistry, it becomes plausible that accurate rate data for most butanol specific reactions based on validated experiments and theory is still not available. For future refinement of kinetic models especially at high temperatures and high pressures accurate data for these reactions is intently needed. For temperatures below 1000 K, one of the important OH consumption reaction which is sensitive to ignition delays is the reaction n-C_4H_9OH + OH reactions for which accurate rate data is also desired.

In addition to the deviations from simulations and experiments above 1000 K as discussed above, the deviations at low temperatures are even more significant as depicted in Fig. 5. One probable reason for this is that the low temperature oxidation mechanism for novel fuels such as butanol is still widely unexplored. At low temperatures below 1000 K, the fuel specific reactions become dominant because of the role of peroxy radicals and their competing decomposition, isomerization and stabilization reactions control the branching leading to auto-ignition. In order to investigate the fuel rich low temperature ignition, our approach is to design new experiments in order to isolate the low temperature sub-mechanism (in a similar fashion as in Ref. [18]), study it in separate experiments and later incorporate the validated sub-mechanism in the comprehensive models. The discrepancies between current "global mechanisms", in predicting ignition delays of novel fuels at high pressures for both low and high temperatures clearly identifies the need for reliable and accurate kinetic data under wide temperature and pressure conditions and that low temperature pressure effects [18–21] which are not studied even for some conventional fuels could be important.

Furthermore, pre-ignition might affect the ignition delay in experiments [12,17] and could be a cause for larger discrepancies at low temperatures. As seen in Fig. 5 the difference between simulation and experiment is significantly larger for temperatures below 1000 K than for higher temperatures. When pre-ignition occurs, the energy release from local combustion areas will increase the temperature and pressure of the unburned mixture [17]. Thus, the additional pressure and temperature gradient cause shorter ignition delay times. An indication, that pre-ignition occurs in our experiments, is shown in Fig. 2. Regarding the pressure and species-emission history of the butanol–air experiment, an increase of these signals can be observed much before the main ignition. In simulation, pressure increases are indicated only shortly before the main ignition. However, this increase is weaker than the one observed in experiment. These effects could be important to the observed discrepancies between simulation and experiment.

5. Conclusions

The ignition delays of *n*-butanol have been investigated in a high-pressure shock tube. Pressure and emission measurements were used to determine the ignition delay times. Experiments behind the reflected shocks were performed at pressures and temperatures ranging from 10 to 42 bar and 770 to 1250 K, respectively. For high temperatures (>1000 K) the experimental and simulation results show an Arrhenius type behavior while for lower temperatures the experimental ignition delay times are significantly shorter than that of the simulation for isochoric conditions. One probable reason for this effect is the pressure and temperature gradients which are caused by shock attenuation. Simulations for comparing the experimental results were performed using two recent comprehensive mechanisms. A similar behavior of the simulation and experimental results is observed, if these gradients are included in the simulation. Nevertheless differences persist between experiment and simulation for higher pressures which could be attributed to a lack of a more detailed low temperature kinetic oxidation mechanism of butanol. However, a good agreement between the experiments and models is observed at 10 bar. The reasons for discrepancies between our experiments and the models at the highest investigated pressures are unknown and difficult to resolve. This paper discusses the role of sensitive reactions at three different temperatures. A sensitivity analysis reveals the relevance of the abstraction reaction of butanol + HO_2 under the conditions of the present study. Comparing the rate coefficients from the two mechanisms adopted here, the rate coefficient for this reaction is about an order of magnitude different in both these mechanisms. An accurate rate coefficient for this reaction based on detailed theory may help improve the agreement of experiments and simulations to some extent. It is important to note, that there could be other sensitive reactions for which accurate kinetic data is needed especially at high pressures to account for the deviations. This demonstrates the need for further investigations concerning the ignition studies of butanol especially in the intermediate and low temperature regime at higher pressures.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.052.

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